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c0106 # Immunosuppressive Medications in Kidney Transplantation

Pitchaphon Nissaisorakarn, Karl L. Womer, Leonardo V. Riella

s0010 INTRODUCTION

p0010 Much of the success of kidney transplantation is due to advances in immunosuppressive medications used during the induction and maintenance phases and for treatment of acute rejection.[1] The term *maintenance immunosuppression* is usually used to describe drug regimens consisting of small-molecule agents administered to stable kidney transplant recipients to prevent transplant rejection. Increasingly, physicians outside specialist transplant centers are involved in the care of kidney transplant recipients. In contrast, biologic agents that are used as induction immunosuppression before the maintenance stage or for the treatment of acute rejection episodes are usually managed by specialist transplant physicians.

s0015 SMALL-MOLECULE DRUGS

s0020 Corticosteroids

p0015 Corticosteroids have been a cornerstone of transplant immunosuppression for the past 50 years, both as maintenance immunosuppression and for treatment of acute rejection.

s0025 Mechanism of Action

p0020 Corticosteroids suppress the production of numerous cytokines and vasoactive substances, including interleukin (IL)-1, IL-2, tumor necrosis factor (TNF)-α, major histocompatibility complex class II, chemokines, and proteases. Corticosteroids also cause neutrophilia (often with a left shift), but neutrophil chemotaxis and adhesion are inhibited. Corticosteroids act as agonists of glucocorticoid receptors but at higher doses have receptor-independent effects. Corticosteroid receptors (CRs) belong to a family of ligand-regulated transcription factors called *nuclear receptors* and are generally present in the cytoplasm in an inactive complex with heat shock proteins. The binding of corticosteroids to the CRs dissociates heat shock protein and forms the active corticosteroid-CR complex, which migrates to the nucleus and dimerizes on palindromic DNA sequences in many genes; this action is called the *corticosteroid response element*. The binding of CR in the target genes' promoter region can lead to either induction or suppression of gene transcripts (e.g., of cytokines). CRs also exert effects by interacting directly with other transcription factors independent of DNA binding. Corticosteroids influence immune responses by regulating the transcription factors activator protein 1 (AP-1) and nuclear factor-κB (NF-κB). Typically, NF-κB is present as an inactive complex with an inhibitor of NF-κB (IκB), but it can be released by IκB kinase. Corticosteroids limit inflammation by stimulating IκB, which then competes with the IκB–NF-κB complex for degradation by IκB kinase. Corticosteroids also stimulate lipocortin, which inhibits phospholipase A_2, thereby inhibiting the production of leukotrienes and prostaglandins. Also, animal data suggest that there may be direct corticosteroid effects on kidney glomerular cells separate from its immunosuppressive effects.[2] The net effect of corticosteroids involves immunosuppressive effects on cytokines, adhesion molecules, apoptosis, and inflammatory cells, as well as possible direct effects on glomerular cells.

Pharmacokinetics s0030

The major corticosteroids used are oral prednisone (or prednisolone) p0025
and oral or intravenous (IV) methylprednisolone. The oral agents have good bioavailability and short pharmacokinetic half-lives (60–180 minutes) but long biologic half-lives (18–36 hours). Corticosteroids are eliminated by hepatic conjugation and are excreted by the kidneys as inactive metabolites. Coadministration of inducers of cytochrome P-450 enzymes that metabolize corticosteroids (e.g., phenytoin, rifampin) decreases corticosteroid half-life, whereas concomitant use of P-450 3A4 inhibitors (e.g., ketoconazole) has the opposite effect. Because corticosteroid levels are not routinely monitored, dosage adjustment during concurrent therapy with these medications is problematic. For treatment of acute rejection, pulse doses of 250 to 500 mg of methylprednisolone are typically used.

Side Effects s0035

Side effects of corticosteroid therapy are common and associated with p0030
significant morbidity, particularly cataracts, osteoporosis, and avascular necrosis, especially of the femoral head (Table 106.1). Other side effects include hypertension, increased appetite with weight gain, hyperglycemia, hyperlipidemia, cushingoid features, acne, skin striae, psychiatric disturbances, sleep disorders, peptic ulcer disease, pancreatitis, colonic perforation, growth retardation, and myopathy. Infection risk is also increased and is excessive if high-dose therapy is prolonged. Interestingly, corticosteroids are not associated with an increased incidence of malignancy. Although corticosteroids are generally considered safe in pregnancy, fetal adrenal suppression has been reported. Rapid corticosteroid reduction protocols with low maintenance doses (5–10 mg/day of prednisone) can improve corticosteroid tolerability by ameliorating many of these side effects. Withdrawing corticosteroids altogether is another strategy that has been implemented to minimize these side effects, especially in selected low-immunologic risk patients.[3] This was demonstrated by a recent trial comparing rabbit antithymocyte globulin (ATG) to basiliximab induction therapy with rapid steroid withdrawal. Not only was rabbit ATG nonsuperior to basiliximab for the prevention of biopsy-proven acute rejection after steroid withdrawal within 1 year after kidney transplantation, for patients with low immunologic risk, rapid steroid withdrawal was achieved without loss of efficacy and with lower risk of post-transplant diabetes.[4] However, most transplant centers advocate for low-dose maintenance steroids, especially in those at high risk for adverse outcomes, such as those highly sensitized and those who may experience delayed graft function, in

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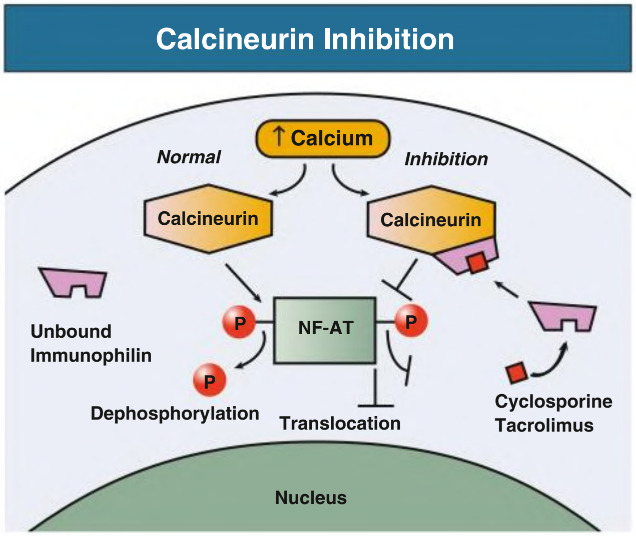
TABLE 106.1 Side Effects of Small-Molecule Immunosuppressive Medications

	Cyclosporine	Tacrolimus	Mycophenolate	Azathioprine	Corticosteroids	mTOR Inhibitors	Leflunomide
Kidney	Nephrotoxicity, type 4 RTA, HTN, diuretic resistance, hyperkalemia, hypomagnesemia, hypophosphatemia	Nephrotoxicity, type 4 RTA, HTN, diuretic resistance, hyperkalemia, hypomagnesemia, hypophosphatemia			HTN, hypokalemia, diuretic resistance	Synergistic nephrotoxicity with CNIs, delayed recovery from ATN, proteinuria, hypokalemia, HTN	
Gastrointestinal		Diarrhea, abdominal pain	Diarrhea, nausea and vomiting, gastritis, esophagitis, oral and colonic ulcers	Nausea and vomiting, hepatotoxicity, pancreatitis	Peptic ulcers, gastritis, esophagitis, diarrhea, colonic perforation	Diarrhea	Nausea, diarrhea, hepatitis
Hematologic	Thrombotic microangiopathy	Thrombotic microangiopathy	Anemia, leukopenia, thrombocytopenia	Anemia, leukopenia, thrombocytopenia	Leukocytosis, polycythemia	Thrombotic microangiopathy, anemia, thrombocytopenia	Anemia, leukopenia
Metabolic	Hyperlipidemia, hyperuricemia, gout, glucose intolerance	New-onset diabetes			Hyperlipidemia, hyperuricemia, hyperglycemia, osteoporosis, avascular necrosis, increased appetite and weight gain	Hyperlipidemia	
Dermatologic	Gingival hyperplasia, coarsened facial features	Alopecia			Hirsutism, acne, cushingoid facies, buffalo hump	Impaired wound healing, oral ulcers	Alopecia, rash
Neurologic	Encephalopathy, insomnia, myopathy, tremors	Encephalopathy, insomnia, myopathy, tremors	Progressive multifocal leukoencephalopathy		Psychosis, insomnia, myopathy	Reflex sympathetic dystrophy	
Other	Edema	Myocardial hypertrophy	Viral infections, pulmonary edema in elderly		Cataracts	Lymphocele, interstitial pneumonitis, rash, edema	

ATN, Acute tubular necrosis; *CNIs,* calcineurin inhibitors; *HTN,* hypertension; *mTOR,* mammalian target of rapamycin; *RTA,* renal tubular acidosis.



f0010 **Fig. 106.1 Calcineurin Inhibition.** During normal T-cell activation, calcium release activates calcineurin's phosphatase activity, causing dephosphorylation of the transcription factor, nuclear factor of activated T cells (NF-AT), and subsequent translocation to the nucleus. Cyclosporine and tacrolimus form a complex with immunophilins (cyclophilin or FK-binding protein 12, respectively), which bind calcineurin and sterically inhibit the phosphatase activity, preventing dephosphorylation and nuclear translocation of NF-AT.

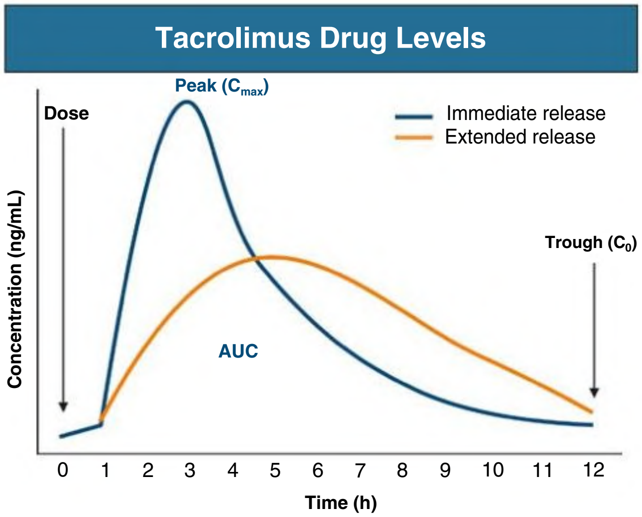


Fig. 106.2 Drug Levels During a Dosing Interval Using Either Immediate- or Extended-Release Tacrolimus. The drug concentration is lowest just before the dose is taken (C_0), then rises to a peak concentration at a certain time after the dose (C_{max}). The area under the concentration-time curve (AUC) describes total drug exposure during the entire administration interval. f0015

which early glucocorticoid withdrawal has been associated with an increased risk of allograft loss.[5]

s0040 Calcineurin Inhibitors

p0035 Calcineurin inhibitors (CNIs), including cyclosporine and tacrolimus, are fungus-derived lipid-soluble small molecules that are the mainstay of current maintenance immunosuppression. Considerable variability exists in their pharmacokinetics, interactions, and side effect profiles, which raises the question of which agent to use in the individual patient. Cyclosporine is a cyclic, lipophilic undecapeptide with several *N*-methylated amino acids, which may explain its resistance to inactivation in the gastrointestinal (GI) tract. Tacrolimus is a macrolide lactone antibiotic.

s0045 Mechanism of Action

p0040 CNIs exert their effect by binding to cytoplasmic proteins called *immunophilins* (Fig. 106.1). Cyclosporine binds to cyclophilin, and tacrolimus binds to FK-binding protein 12 (FKBP12).[6] Such binding enhances immunophilin affinity for and subsequent inhibition of calcineurin, which is a calmodulin-activated serine phosphatase important for dephosphorylation of inactive nuclear factor of activated T cells (NF-AT). Nuclear translocation of dephosphorylated (active) NF-AT, in association with other transcription factors, initiates downstream events, leading to T-cell activation. The tacrolimus-FKBP12 complex inhibits calcineurin with greater molar potency than the corresponding cyclosporine complex. Cyclosporine and tacrolimus can interfere with calcineurin's activation on substrates other than NF-AT, which likely explains many of their side effects. Treatment with CNIs also causes upregulation of the cytokine transforming growth factor-β (TGF-β), which has significant immunosuppressive properties but also promotes the production of matrix proteins and tissue fibrosis. CNIs can suppress the immune response by calcineurin-independent pathways, likely by blocking intracellular signaling pathways specific to T cells. These agents' ability to interfere with two distinct mechanisms of T-cell activation contributes to their highly specific immunosuppressive properties.

Pharmacokinetics, Monitoring, and Drug Interactions s0050

After a dose of CNI, there is an initial absorptive phase, during which p0045
blood concentrations reach a peak level (C_{max}).[7] Typically, C_{max} occurs during the first 2 to 3 hours after the dose and corresponds to the time of maximum calcineurin inhibition. Drug levels then fall because of metabolism (also known as the *elimination phase*) until they are at the trough level (C_0) immediately before the next dose. The total drug exposure throughout the period from one dose until the next is the area under the concentration-time curve (AUC; Fig. 106.2).

For both CNIs, most of the interpatient and intrapatient vari- p0050
ability occurs in the absorption rather than in the elimination phase. The oil-based formulation of cyclosporine requires solubilization in bile and is plagued by highly variable and unpredictable bioavailability. The microemulsion preparation of cyclosporine (modified) has enhanced bioavailability and less dependence on bile secretion. In blood, cyclosporine resides primarily in erythrocytes (60%–70%) and leukocytes, with some binding to lipoproteins and, to a lesser extent, other plasma proteins. Cyclosporine is metabolized primarily by CYP3A4, a member of the cytochrome P450 superfamily. Metabolism occurs mostly in the liver. Interindividual differences in CYP3A4 activity and the large number of exogenous and endogenous substances capable of altering its function and expression explain the wide variation in clearance rates. One factor is the multidrug resistance 1 gene product, P-glycoprotein, which is variably expressed in the intestine and reduces the absorption of several xenobiotics, including CNIs, by transport out of intestinal epithelial cells. The average half-life of cyclosporine is about 19 hours, with excretion primarily in bile. Generic formulations of modified cyclosporine exist, although they may not have identical pharmacokinetics and therefore may not be readily interchangeable.

The absorption of tacrolimus, like that of cyclosporine, is highly vari- p0055
able, with bioavailability ranging from 5% to 67%. Absorption is not bile dependent but does depend on GI transit time and is affected by the presence or absence of food and the lipid content of food. Clearance appears to be faster in children, necessitating higher or more frequent dosing. In the United States, Black and Hispanic patients also require higher doses than White patients to achieve equivalent therapeutic

levels, likely because of genetic differences in subtypes of CYP3A, such as CYP3A5 (faster metabolism of CNIs).[8] Similarly, higher tacrolimus dose requirements were reported for non-White Brazilian transplant recipients and for Black transplant recipients in the United Kingdom.[9,10]

p0060 In blood, tacrolimus distributes primarily to erythrocytes, with whole-blood concentrations 10 to 30 times higher than in plasma. In contrast to cyclosporine, no lipoprotein binding occurs. Tacrolimus has 20- to 30-fold higher potency than cyclosporine on a molecular weight basis. Like cyclosporine, metabolism occurs through the CYP3A4 system, with pharmacokinetics also affected by intestinal P-glycoprotein. Both CNIs are generally administered twice daily, but newer slow-release formulations for tacrolimus are now available with once-a-day dosing and reduced peak levels, with potential benefits in lowering side effects such as tremor (reduced peak) and improvement in compliance (see Fig. 106.2). Because of the narrow therapeutic index for CNIs, the variability of concentrations among patients after a dose, and the potential for drug interactions, monitoring of drug levels is required to ensure both safety and adequacy.

p0065 Given the complementary influence of both CYP3A4 and P-glycoprotein on the pharmacokinetic profiles of CNIs, it is assumed that CNI-drug interactions on both CYP3A4 and P-glycoprotein have similar complementary effects on CNI pharmacokinetics. Drugs that competitively inhibit CYP3A4 activity, such as ketoconazole, usually also inhibit P-glycoprotein, thereby increasing the bioavailability of CNIs, with potential for toxicity. Likewise, drugs such as phenytoin that increase CYP3A4 levels tend to upregulate P-glycoprotein, decreasing overall bioavailability. In this case the likelihood of rejection increases. Diarrhea may increase the absorption of CNIs by affecting P-glycoprotein transport in the gut, and monitoring of CNI levels is recommended in the setting of persistent diarrhea for more than 3 days. Despite similar drug interactions, the age-related and ethnic differences in pharmacokinetics between the two CNIs are likely to influence the degree and importance of such interactions. Common interactions are presented in Box 106.1.

s0055 Side Effects

p0070 Cyclosporine and tacrolimus have similarities and differences in their toxicity profiles (see Table 106.1). Both can cause nephrotoxicity, hyperkalemia, hyperuricemia, hypomagnesemia and hypophosphatemia (secondary to urinary loss), type 4 renal tubular acidosis, hypertension, diabetes, and neurotoxicity. Gingival hyperplasia, hirsutism, hypertension, hyperuricemia, and hyperlipidemia are more common with cyclosporine, whereas tremor and glucose intolerance are more common with tacrolimus. The most common and vexing problem with CNIs is nephrotoxicity, with its importance evident from heart and liver transplant recipients, in whom CNIs were associated with progression to end-stage kidney disease. Both reversible hemodynamic and irreversible structural components underlie the nephrotoxicity of CNIs. Reversible vasoconstriction is caused by direct vascular effects but is also because of activation of the renin-angiotensin system (RAS), endothelin, thromboxane, and the sympathetic nervous system. Over time, chronic kidney injury occurs, characterized by afferent arterial hyalinosis and tubulointerstitial fibrosis, presumably as the result of prolonged kidney vasoconstriction with ischemia and direct tubular toxicity. Finally, CNIs can cause thrombotic microangiopathy (TMA) in a minority of cases, probably by direct endothelial cell injury and dysfunction.

s0060 Mycophenolate

p0075 Mycophenolate mofetil (MMF) and enteric-coated mycophenolate sodium (EC-MPS) are important components of immunosuppressive regimens that are associated with some of the most successful outcomes in kidney transplantation.[11] Because of its well-documented efficacy and acceptable side effect profile, MMF is by far the most frequently used antiproliferative agent and is commonly used in combination with CNIs.

BOX 106.1 Drugs and Other Substances That Interact With Calcineurin Inhibitors

Increase Blood Levels (P450-3A4 and/or P-Glycoprotein Inhibitors) b0010

- Ketoconazole
- Fluconazole
- Itraconazole
- Voriconazole
- Erythromycin
- Clarithromycin
- Diltiazem
- Verapamil
- Nicardipine
- Cimetidine
- Methylprednisolone
- Metronidazole
- Ezetimibe
- Metoclopramide
- Fluvoxamine
- Human immunodeficiency virus (HIV) protease inhibitors
- Lovastatin
- Atorvastatin
- Simvastatin
- Grapefruit juice
- Chamomile
- Wild cherry

Decrease Blood Levels (P450-3A4 and/or P-Glycoprotein Inducers)

- Rifampin
- Rifabutin
- Phenytoin
- Carbamazepine
- Phenobarbital
- Caspofungin
- St. John's wort

Mechanism of Action s0065

The immunosuppressive effects of mycophenolate are likely mediated p0080 through the active metabolite mycophenolic acid (MPA). MMF is a morpholinoethyl ester of MPA, a potent reversible inhibitor of inosine monophosphate dehydrogenase (IMPDH), isoform 2. EC-MPS is a salt that combines an acid (MPA) with a base (sodium). The enteric coating delays MPS release so that the MPA is absorbed in the small intestine rather than in the stomach. MPA noncompetitively inhibits IMPDH, which is the rate-limiting enzyme in the de novo synthesis of guanosine monophosphate (GMP). Inhibition of IMPDH creates a relative deficiency of GMP and a relative excess of adenosine monophosphate (AMP). GMP and AMP levels act as a control on de novo purine biosynthesis; therefore MPA, by inhibiting IMPDH, blocks de novo purine synthesis, which selectively interferes with proliferative responses of T and B cells. Some other cell types, such as GI epithelial cells, use the de novo pathway. Thus, MPA may inhibit replication of GI epithelial cells, leading to the disruption of fluid absorption and diarrhea. However, most other cell types depend primarily on the alternative pathway for DNA synthesis and cell division and thus are relatively spared from toxicity.

s0070 Pharmacokinetics

p0085 MMF is absorbed rapidly and completely from the GI tract and undergoes extensive presystemic deesterification to become MPA, the active form. Food intake can delay the rate of MMF absorption but does not affect the extent. However, coadministration of antacids (proton pump inhibitors), calcium, or cholestyramine decreases absorption by 20% to 40%. EC-MPS has shown bioavailability equivalence and similar efficacy to that of MMF. MPA undergoes enterohepatic circulation, and its plasma concentration shows a secondary peak at 6 to 12 hours after oral administration. The mean contribution of enterohepatic circulation to the overall AUC of MPA is 37% (range, 10%–61%). Most MPA is metabolized in the liver through a phase II glucuronidation process.

p0090 The major metabolite of MPA is the pharmacologically inactive 7-*O*-glucuronide metabolite (MPAG), although two other metabolites, MPA-acylglucuronide (AcMPAG) and MPA-phenyl-glucoside (glucoside-MPA), are isolated from the plasma of kidney transplant patients. AcMPAG has shown in vitro pharmacologic activity (inhibition of IMPDH) and proinflammatory effects and potentially is responsible for the GI toxicity of MPA. The glucuronide metabolites are excreted into the bile, a process mediated by the multidrug resistance–related protein 2 (MRPR2), then undergo deglucuronidation back to MPA by enzymes that are produced by colonic bacteria. Blockade of MRPR2 by an inhibitor, such as cyclosporine but not tacrolimus, decreases the biliary excretion of MPAG and increases plasma MPAG levels. This eventually leads to lower plasma levels of MPA because the glucuronide metabolites no longer can be reabsorbed as MPA by enterohepatic cycling. Thus, tacrolimus-treated patients have higher exposure to MPA than cyclosporine-treated patients. The ultimate elimination pathway for the glucuronide metabolites is through the kidney, and more than 95% of an administered dose eventually is found in the urine as glucuronide metabolites. For the most part, MMF and EC-MPS are used in fixed-dose regimens. No studies in transplantation have shown that monitoring MPA drug AUC or trough levels reduces the rejection rate or minimizes toxicity.

s0075 Side Effects

p0095 MMF and EC-MPS have similar side effect profiles, including GI toxicity, bone marrow suppression, and increased infections, especially those of viral cause (see Table 106.1). GI disturbances include oral ulcerations, esophagitis, gastritis, nausea, vomiting, diarrhea, and colonic ulcers. Diarrhea and leukopenia frequently necessitate a dose reduction that could precipitate allograft rejection. Because the metabolites of MPA appear to play a major role in the GI disturbances associated with MMF, there is little rationale for enteric coating of the prodrug to reduce these symptoms. In fact, randomized controlled trials (RCTs) have found no significant difference in GI adverse events between MMF and EC-MPS.[12–14] MMF is generally avoided during pregnancy because of its teratogenicity in experimental animal models and clinical reports of major fetal malformations.

s0080 Azathioprine

p0100 The use of azathioprine has decreased dramatically in kidney transplantation since the introduction of MMF. Azathioprine is metabolized in the liver to 6-mercaptopurine and further converted to the active metabolite thioinosinic acid by hypoxanthine-guanine phosphoribosyltransferase. Because allopurinol (a xanthine oxidase inhibitor) increases levels of thioinosinic acid and can lead to life-threatening bone marrow suppression, the dose of azathioprine must be substantially reduced or azathioprine substituted with another agent—commonly MMF—in patients taking allopurinol. Azathioprine suppresses the proliferation of activated T and B cells and reduces the number of circulating monocytes by arresting the cell cycle of promyelocytes in the bone marrow. The antiproliferative effect is mediated by the metabolites of azathioprine, including 6-mercaptopurine, 6-thiouric acid, 6-methylmercaptopurine, and 6-thioguanine. These compounds are incorporated into replicating DNA and halt replication. They also block the de novo pathway of purine synthesis by formation of thioinosinic acid; this effect confers specificity of action on lymphocytes that lack a salvage pathway for purine synthesis. The major side effect of azathioprine is bone marrow suppression, leading to leukopenia, thrombocytopenia, and anemia (see Table 106.1), with potentially life-threatening complications arising from the concurrent use of azathioprine and xanthine oxidase inhibitors such as allopurinol. The mean cell volume is commonly increased in patients taking azathioprine, and red cell aplasia occasionally can occur. The hematologic side effects are dose-related and usually reversible on dose reduction or temporary discontinuation of the drug. Other common side effects are increased risk for malignancy (especially skin cancers), hepatotoxicity, pancreatitis, and hair loss.

Mammalian Target of Rapamycin Inhibitors s0085

Mammalian target of rapamycin (mTOR) inhibitors are proliferation p0105
signal inhibitors with immunosuppressive activity. Sirolimus, also known as *rapamycin*, was the first mTOR inhibitor used in transplantation and is a macrolide product of a soil fungus discovered on Easter Island.[15] Everolimus is a rapamycin analog with a similar mechanism of action, immunosuppressive properties, and side effect profile. Although initially used in drug regimens with the intent of minimizing exposure to CNIs with its known side effects, the mTOR inhibitors have been associated with their own set of toxicities that have prevented their widespread use. A recent study failed to demonstrate noninferiority of everolimus compared with MPA in groups with similar tacrolimus or cyclosporine exposure in terms of mean estimated glomerular filtration rate (eGFR) 12 months after transplant.[16] Many trials using mTOR inhibitors in place of CNIs revealed a higher risk of kidney transplant rejection and variable improvement in kidney function.[17–21] Lastly, trials using mTOR inhibitor in place of MMF have suggested a lower risk of opportunistic viral infections, such as cytomegalovirus.[22]

Mechanism of Action s0090

Sirolimus has a structural similarity to tacrolimus and binds to FKBP12. p0110
The affinity of sirolimus for FKBP12 is higher than that of everolimus. mTOR inhibitors do not inhibit calcineurin or the calcium-dependent activation of cytokine genes but instead inhibit cytokine receptor–mediated signal transduction and cell proliferation that block lymphocyte responses to cytokines and growth factors. The sirolimus-FKBP12 or everolimus-FKBP12 complex binds with high affinity to the kinase enzyme mTOR, which is a serine-threonine kinase of the phosphatidylinositol 3–kinase pathway that acts during costimulatory and cytokine-driven pathways. mTOR inhibits a translation repressor protein (4E-BP1) and activates a ribosomal enzyme (p70-S6 kinase), both of which are important for translation of the mRNAs for certain proteins needed for progression from the G1 phase to the S phase of DNA synthesis. mTOR has been identified as the principal controller of cell growth and proliferation. The sirolimus-FKBP12 complex inhibits mTOR-mediated signal transduction pathways by blocking postreceptor immune responses to costimulatory signal 2 during G0 to G1 transition and to cytokine signaling during G1 progression. It also inhibits IL-2– and IL-4–dependent proliferation of T and B cells, leading to suppression of new ribosomal protein synthesis and arrest of the G1-S phase of the cell cycle. Proliferation of nonimmune cells, such as fibroblasts, endothelial cells, hepatocytes, and smooth muscle cells, is also impaired by inhibition of the growth factor–mediated responses (e.g., basic fibroblast growth factor, platelet-derived growth factor, vascular

endothelial cell growth factor, and TGF-β). In addition, mTOR contributes to several protein synthesis pathways that could be involved in oncogenesis.

s0095 ### Pharmacokinetics

p0115 The oral bioavailability of sirolimus is poor (10%–16%), with significant interindividual and intraindividual variability. Peak concentrations occur approximately 1 to 2 hours after an oral dose, and sirolimus distributes extensively into tissues, including blood cells. The oral bioavailability of everolimus is higher than that of sirolimus. High-fat meals increase sirolimus levels while decreasing everolimus levels. Because sirolimus has a relatively long half-life (~62 hours), it is reasonable to wait 1 week (approximately three half-lives to achieve steady state) before monitoring sirolimus blood levels after initiation or dose adjustment. Sirolimus is metabolized by the P-450 3A4 isoenzyme and P-glycoprotein system and thus has interactions similar to those described for CNIs (see Box 106.1).

p0120 When an mTOR inhibitor is administered simultaneously with cyclosporine, the C_{max} and AUC for both compounds are increased. Also, cyclosporine clearance may be reduced during concurrent therapy.

s0100 ### Side Effects

p0125 The mTOR inhibitors have a wide variety of toxicities (see Table 106.1). The most common adverse effects associated with sirolimus are dose-dependent hypertriglyceridemia, anemia, thrombocytopenia, and leukopenia. Other adverse effects include impaired wound healing and dehiscence, formation of lymphoceles, oral ulcers, reduced testosterone levels, proteinuria, diarrhea, and pneumonitis. Although not inherently nephrotoxic, the mTOR inhibitors result in kidney graft damage through several mechanisms. When used in combination with full-dose cyclosporine (and probably tacrolimus), sirolimus potentiates CNI-induced nephrotoxicity. In patients with kidney impairment, sirolimus is associated with marked yet potentially reversible proteinuria and worsening of established proteinuria. Sirolimus also can cause delayed recovery from acute tubular necrosis and may be linked to podocyte injury and focal segmental glomerulosclerosis. Finally, cases of TMA have been reported, and there is concern that higher doses of sirolimus may inhibit endothelial cell growth. Interestingly, sirolimus-based regimens have been associated with a reduced incidence of post-transplantation malignancy. Some physicians regard sirolimus as the preferred immunosuppressive agent in transplant patients who develop malignancy because of data suggesting that mTOR inhibitors may reduce the risk of malignancies[23,24]; however, most data are limited to kidney transplant recipients with squamous cell skin cancer.[25,26] Sirolimus is not routinely used during pregnancy because of its teratogenicity in animal models, although successful pregnancies have been reported.

s0105 ## BIOLOGIC AGENTS

p0130 Biologic agents in the form of polyclonal antibodies and monoclonal antibodies (mAbs) are frequently used in kidney transplantation either as induction therapy or for the treatment of rejection. Polyclonal antibodies are derived from horses or rabbits; historically, mAbs have been murine in origin. However, because foreign proteins can elicit an immune response, there has been an attempt to replace murine monoclonal products with humanized or chimeric mAbs (Fig. 106.3). Humanized antibodies are produced by merging the DNA that encodes the antigen-binding portion of a monoclonal mouse antibody with human antibody–producing DNA. Mouse hybridomas are then used to express this DNA to produce hybrid antibodies that are not as immunogenic as the murine variety. Chimeric antibodies use the same

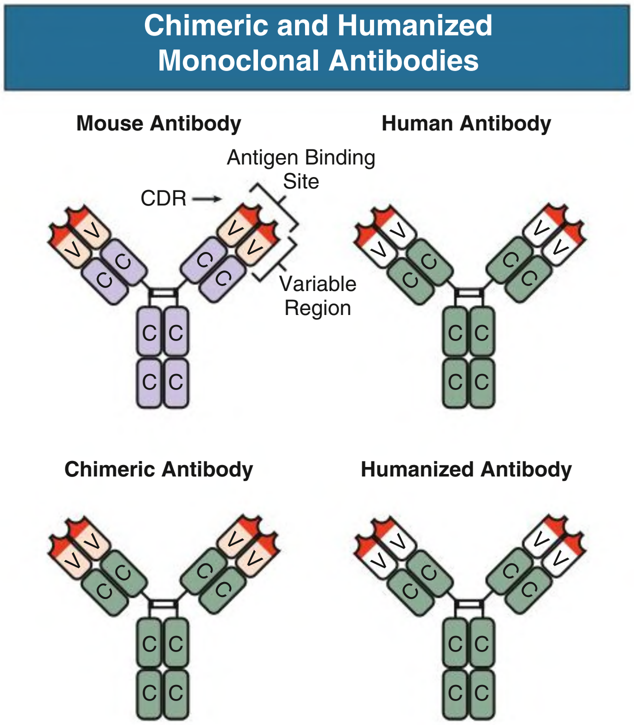


Fig. 106.3 Chimeric and Humanized Antibodies. Chimeric antibodies consist of human constant *(C)* regions and mouse variable *(V)* regions. A chimeric antibody therefore retains the antigen binding site of the mouse antibody but with fewer amino acid sequences foreign to the human immune system than a standard mouse antibody. Humanized monoclonal antibodies retain only the minimum necessary parts of the mouse antibody for antigen binding, the complementarity-determining region (CDR, highlighted in *red*), and therefore are even less immunogenic in the human host. f0020

strategy but for the entire variable region and thus are more immunogenic than humanized antibodies. Polyclonal antibodies and mAbs can be depleting agents or immune modulators.

Polyclonal Antilymphocyte Sera s0110

Polyclonal antilymphocyte agents are produced by immunizing animals with human thymus–derived lymphoid cells. Although rabbit ATG is currently the preferred preparation, equine preparations historically have been used. Most regimens involve daily IV administration of ATG for 3 to 4 days either as induction therapy or for treatment of corticosteroid-resistant rejection. ATG contains antibodies that react against a variety of targets, including red blood cells, neutrophils, dendritic cells, and platelets. ATG binds to multiple epitopes on the surface of T cells and induces a rapid lymphocytopenia by several mechanisms, including complement-dependent cytolysis, cell-dependent phagocytosis, and apoptosis. ATG is a potent immunosuppressive, and T- and B-lymphocyte counts can remain depressed up to 24 hours after administration. The lack of specificity coupled with marked immunosuppression increases the risk for infection and cancer. Because polyclonal agents are xenogeneic proteins, they may elicit a number of side effects, including fever and chills. The initial lysis and activation of T cells that follow ATG administration may generate significant side effects after the first dose with the release of TNF-α, interferon-γ (IFN-γ), and other cytokines. Therefore, steroids are frequently used before ATG infusion. Less commonly, ATG can induce a serum sickness–like syndrome and, rarely, acute respiratory distress syndrome. p0135

s0115 Humanized Monoclonal Anti-CD52 Antibody

p0140 Alemtuzumab is a humanized IgG1 mAb directed against CD52, a glycoprotein found on circulating T and B cells, monocytes, macrophages, natural killer cells, and granulocytes. Alemtuzumab was originally approved for the treatment of B-cell chronic lymphocytic leukemia, but it has been used off label as an induction agent in kidney transplantation. Treatment results in a rapid and effective depletion of peripheral and central lymphoid cells, which may take months to return to pretransplantation levels. Side effects of alemtuzumab include first-dose reactions, neutropenia, anemia, and rarely, pancytopenia and autoimmunity (e.g., hemolytic anemia, thrombocytopenia, and hyperthyroidism). The risks for immunodeficiency complications such as infection and malignant neoplasia with alemtuzumab appear to be similar to ATG.

s0120 Monoclonal Anti-CD25 Antibody

p0145 The alpha subunit of the IL-2 receptor (CD25) is upregulated on activated T cells and leads to the expression of high-affinity IL-2 receptors. The engagement of IL-2 receptors by IL-2 triggers the activated T cell to undergo proliferation. Basiliximab is a chimeric mAb with a specificity for CD25; it induces relatively mild immunosuppression and is used as an induction agent to prevent rejection but not to treat established rejection.[27,28] Although the exact mechanism of action is not fully understood, it is clear that significant depletion of T cells does not play a major role. Saturation of the IL-2 receptor alpha subunit persists for up to 25 to 35 days after treatment with basiliximab. Although saturation is important as a determinant of minimal blood concentrations, it is not predictive of rejection. No major side effects have been associated with anti-CD25 induction compared with transplant recipients receiving no anti-CD25 induction.

s0125 B-Cell–Depleting Monoclonal Anti-CD20 Antibody

p0150 Rituximab is an engineered chimeric mAb that contains murine heavy- and light-chain variable regions directed against CD20 plus a human IgG1 constant region.[29] The CD20 antigen, a transmembrane protein, is found on immature and mature B cells and on malignant B cells. CD20 mediates proliferation and differentiation of B cells. Rituximab directly inhibits B-cell proliferation and induces apoptosis and lysis by complement-dependent cytotoxicity, antibody-dependent cell cytotoxicity, and activation of tyrosine kinases as a direct effect of the antibody binding to its CD20 ligand. Rapid and sustained depletion of circulating and tissue-based B cells occurs after IV administration, and recovery does not begin until approximately 6 months after treatment. Although plasma cells are usually CD20 negative, many are short lived and require replacement from CD20-positive precursor cells. In addition, CD20-positive B cells can act as secondary antigen-presenting cells (APCs), thereby enhancing T-cell responses. Thus, by targeting CD20 on precursor B cells, rituximab decreases the production of activated B cells and limits their antibody production and antigen presentation capability. Most adverse events are first-infusion effects, such as fevers and chills, and are generally of mild severity, becoming less frequent with subsequent infusions. Viral infections, including reactivation of hepatitis B virus and JC virus, have been reported, although it is not known whether these events are specific to the agent or instead reflect the overall state of immunosuppression. Antichimeric antibodies develop in some patients, but their true incidence and therapeutic significance are uncertain. Rituximab has been used in kidney transplantation to treat antibody-mediated rejection and in combination with IV immunoglobulin (IVIG) to reduce high-titer anti–human leukocyte antigen (anti-HLA) antibodies in highly sensitized patients awaiting kidney transplantation.[30] Rituximab is also used as induction therapy after desensitization therapy for ABO blood group–incompatible and high-risk positive crossmatch kidney transplantation. Finally, rituximab is often used to treat post-transplantation lymphoproliferative disease. A phase 1b study assessing the safety, pharmokinetics, and pharmacodynamics of obinutuzumab, a novel type 2 anti-CD20 monoclonal antibody in highly sensitized patients with kidney failure, demonstrated good tolerability and effective depletion of B lymphocytes. However, its effect on reductions in anti-HLA alloantibodies were inconsistent and not clinically meaningful.[31]

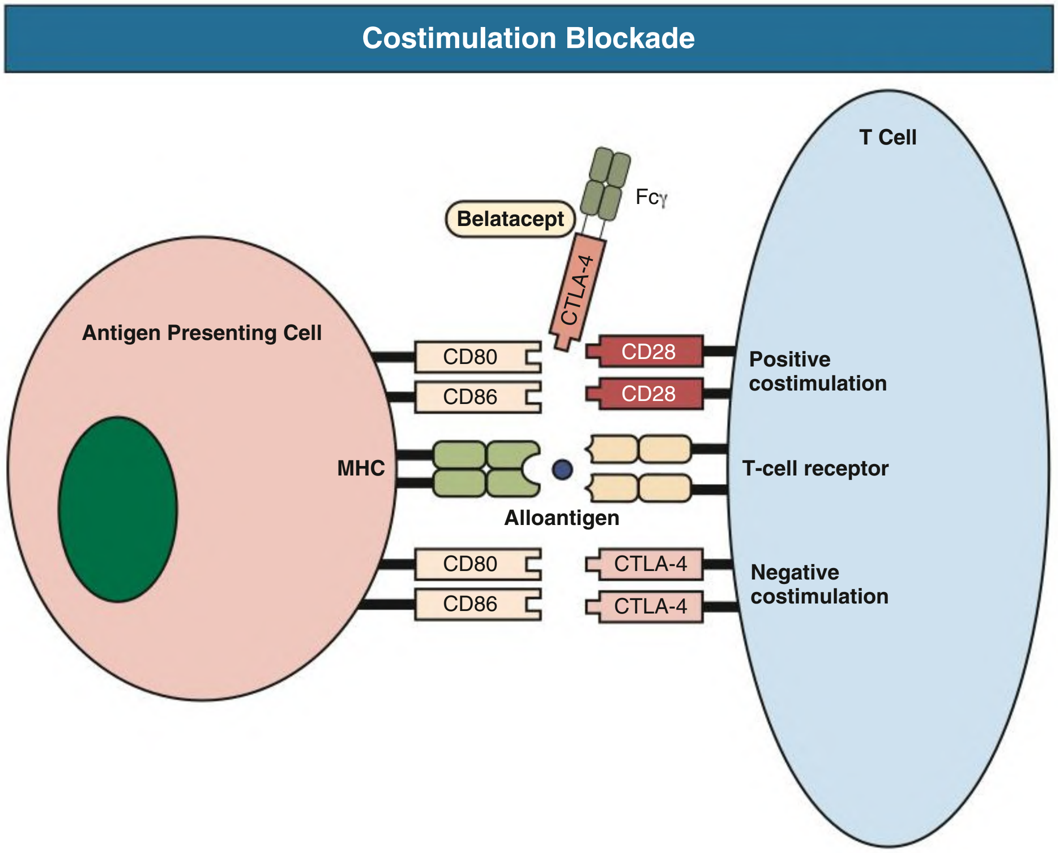
Intravenous Immunoglobulin s0130

IVIG products are known to have powerful immunomodulatory p0155
effects in inflammatory and autoimmune conditions. The mode of action of IVIG is not well understood. In kidney transplantation, the most important effect appears to be a reduction of alloantibodies through inhibition of antibody production and increased catabolism of circulating antibodies. Additional potential mechanisms include inhibition of complement-mediated injury, inhibition of inflammatory cytokine generation, and neutralization of circulating antibodies by anti-idiotypes. Side effects related to IVIG administration include minor self-limited reactions, such as flushing, chills, headache, myalgia, and arthralgia. Rarely, anaphylactic reactions may occur. Delayed reactions include severe headache and aseptic meningitis, which respond to analgesics. More recently, severe thrombotic events have been linked to the administration of IVIG products. Osmotic injury of the proximal tubular epithelium can occur after administration of sucrose-containing IVIG preparations. This tubular injury is self-limited and can be minimized or avoided by use of sucrose-free preparations and by slowing rate of infusion. In combination with plasma exchange, IVIG appears to offer significant benefits in the desensitization of positive-crossmatch and ABO-incompatible patients to allow successful transplantation as well as in the treatment of antibody-mediated rejection. Alone or in combination with rituximab, IVIG has been successful in the desensitization of highly sensitized waitlisted patients to increase the chances of finding a compatible donor.

Belatacept s0135

Costimulation blockade is an immunosuppression alternative for p0160
kidney transplant recipients. Belatacept, a first-in-class costimulation blocker, is a fusion protein that binds to CD80 and CD86 on APCs to prevent T cell activation and proliferation (Fig. 106.4). The drug primarily affects the costimulatory CD28 pathway, preventing full T-cell activation. Belatacept was initially approved by the U.S. Food and Drug Administration (FDA) for de novo kidney transplantation only in Epstein-Barr virus (EBV)-seropositive adult patients because of concerns that administering belatacept might increase the risk for early post-transplant lymphoproliferative disorder (PTLD) in EBV-seronegative patients. Also, it has been used off label for conversion from CNIs after kidney transplantation, in particular in patients intolerant to CNIs. Converting from CNIs to belatacept has been associated with a higher risk of rejection when done in the first year after transplant in sensitized patients; however, there was no difference in rejection-free, patient, or graft survival over 5 years.[32] Other potential benefits of conversion include improved glycemic parameters in those with diabetes[33] and improved kidney function in those with chronic vascular lesions[34] on kidney biopsy. Side effects of belatacept include anemia and leukopenia. Rare but serious side effects include progressive multifocal leukodystrophy and PTLD.

In the phase III BENEFIT trial, belatacept showed a significantly p0165
better GFR despite higher rates of early acute rejection compared with cyclosporine.[35] Belatacept-treated patients also had better blood pressure and lipid control compared with cyclosporine controls. The



f0025 **Fig. 106.4 Costimulation Blockade.** Lymphocyte T-cell activation requires two signals with the first signal mediated by the major histocompatibility complex (MHC) and the T-cell receptor, and the second signal (positive costimulation) mediated by CD80/CD86 on the antigen-presenting cell (APC) and *CD28* on the T cell. Negative costimulation is mediated by the cytotoxic T lymphocyte–associated protein 4 (CTLA-4) on activated T cells that binds to CD80/CD86 on the APC and suppresses T-cell responses (negative costimulation). Belatacept is a CTLA-4 fusion protein that binds CD80/CD86 and blocks positive costimulation via the CD28 pathway, thus preventing T-cell activation.

s0140 7-year follow-up data showed that the composite endpoint of graft
s0145 and patient survival was better with belatacept, but the individual graft
p0175 survival and patient survival were separately not statistically different compared with cyclosporine.[36] Belatacept continued to show better 7-year kidney function with no increased risk for late PTLD in EBV seropositive patients.

p0170 Other studies looked at patients who were not included in the BENEFIT trial. The BENEFIT-EXT trial included patients who received a kidney transplant from an expanded criteria donor, donation after cardiac death donor, or a donor with long cold ischemia time.[37] This trial also found that patient and graft survival with belatacept were not inferior, but unlike in the BENEFIT trial, the acute rejection rate and GFR were similar between belatacept and cyclosporine. More recently a retrospective study of the Scientific Registry of Transplant Recipients database compared the 1-year outcomes between belatacept and tacrolimus.[38] Similar to the BENEFIT trial, this study showed no difference in graft survival, but better kidney function despite a greater risk for acute rejection was seen with belatacept compared with tacrolimus. Also, a lower risk for new-onset diabetes after transplantation was observed with belatacept. In summary, belatacept seems to better preserve GFR and have a better metabolic profile compared with CNIs, but the high acute rejection rate has affected its broader use because of concerns of long-term impact on graft survival.

Other Agents Used in Transplantation

Bortezomib

Bortezomib is an antineoplastic agent originally approved for the treatment of plasma cell dyscrasias such as multiple myeloma and several types of lymphomas. It inhibits proteasomes, which are enzyme complexes that regulate protein homeostasis. Specifically, bortezomib reversibly inhibits chymotrypsin-like activity at the 26S proteasome, leading to activation of signaling cascades, cell-cycle arrest, and apoptosis. It targets mature rapidly proliferating antibody-producing plasma cells but also interferes with T-cell function and IL and TNF production. Bortezomib has been used off label for both primary and refractory antibody-mediated rejection, experimental desensitization protocols, and induction immunosuppression in patients who are highly sensitized with HLA antibodies.[39–41] Also, the BORTEJECT study is an ongoing phase 2 RCT investigating the impact bortezomib might have on treating late antibody-mediated rejection.[42] Bortezomib is administered via IV and is metabolized by the liver. GI symptoms are common, but thrombocytopenia and peripheral neuropathy are significant side effects. Herpes zoster prophylaxis is required.

Eculizumab s0150

Eculizumab is a humanized monoclonal antibody directed against p0180
complement protein C5 that prevents cleavage into C5a and C5b, thus

b0015 **BOX 106.2 Prescribing Considerations for Immunosuppressive Medications**

Tacrolimus

- Close monitoring of tacrolimus level is recommended in patients with diarrhea. Diarrhea elevates tacrolimus levels because of downregulation of (1) intestinal CYP450 and (2) p-glycoprotein, which normally transports drugs back into the intestinal lumen from enterocytes, thus increasing bioavailability.
- Hair loss associated with tacrolimus has been suggested to be dose-related and at times may respond to biotin supplementation; however, if persistent and severe, it may lead to poor compliance and a need for alternate therapy.
- Neurologic effects include night tremors, headaches, and tremors that can sometimes be alleviated with extended-release formulation.
- Urge cautious tacrolimus use with drugs that affect CYP450.
- Hypomagnesemia is common because tacrolimus downregulates TRPM6 and may also increase claudin-14 expression, which inhibits paracellular magnesium transport.
- Monitor for QT prolongation.
- It is associated with thrombotic microangiopathy and posttransplant diabetes.
- There is high interindividual variability on pharmacokinetics dependent on ethnicity; CYP3A5*1 polymorphism is high in African Americans and may lead to a higher tacrolimus dosage requirement.

Cyclosporine

- It is associated with hirsutism and gingival hyperplasia.

Mycophenolate

- Myelosuppression is a common side effect, and close complete blood count (CBC) monitoring is required.
- Gastrointestinal (GI) side effects are common. In cases of significant GI side effects, patients could be tried on enteric-coated mycophenolate mofetil (mycophenolic acid). Spacing out mycophenolic acid from twice daily to four times daily also sometimes helps.
- It is associated with teratogenicity; women of child-bearing age should be on dual contraception while on mycophenolate and should stop mycophenolate at least 6 weeks before conception.

Mammalian target of rapamycin inhibitors (mTORi)

- Routine monitoring of lipid panel and proteinuria is recommended.
- These are associated with pneumonitis.
- There is poor wound healing; these should not be used perioperatively.

Azathioprine

- Azathioprine has been associated with skin cancer with long-term use.
- Avoid use with allopurinol or febuxostat because it may increase serum concentrations of the active metabolite of azathioprine.
- Myelosuppression is a common side effect and close CBC monitoring is required, especially in those with low or absent thiopurine methyltransferase activity, often manifesting as macrocytic anemia.
- It is acceptable for use during pregnancy.

Belatacept

- There is a Black Box warning in Epstein-Barr virus–naive recipients, and it may increase post-transplant lymphoproliferative disorder risk.

blocking the subsequent formation of the terminal complex C5b-9 or membrane attack complex. Eculizumab prevents antibody-dependent complement-mediated cytotoxicity; therefore it has been used as an off-label treatment for antibody-mediated rejection. However, because of its experimental use and extreme cost, eculizumab is usually reserved as a rescue therapy for allografts resistant to other antirejection therapies. Eculizumab use in severe acute antibody-mediated rejection appears to be effective, but long-term follow-up of eculizumab-treated patients did not show prevention of chronic glomerular changes such as transplant glomerulopathy.[43–46] A phase II RCT in living-donor kidney transplant recipients who required desensitization therapy showed that eculizimab did not reduce the risk of biopsy-proven acute antibody mediated rejection, graft loss, or death.[47] As a result of increased incidence of meningococcal infections associated with eculizumab, meningococcal vaccine and antibiotic prophylaxis are required before starting therapy. Finally, eculizimab has been used successfully in the prevention of recurrence of certain complement-mediated glomerular diseases, such as atypical hemolytic syndrome.[48]

See Box 106.2 for a list of considerations when prescribing immunosuppressive medications. p0185

SELF-ASSESSMENT QUESTIONS

o0010 1. What is the *most* common and vexing problem with cyclosporine and tacrolimus in kidney transplantation?
o0015 **A.** Nephrotoxicity
o0020 **B.** Hypertension
o0025 **C.** TMA
o0030 **D.** Alopecia
o0035 2. Belatacept offers patients a CNI alternative to maintenance immunosuppression and works by which mechanism of action?
o0040 **A.** Blocks complement C5 cleavage
o0045 **B.** Blocks CD25 on the IL-2 receptor
o0050 **C.** Blocks CD28-mediated costimulation of T cells
o0055 **D.** Inhibits B-cell proliferation by targeting CD20

3. Which immunosuppressive agent has been associated with teratogenicity in experimental animals and major fetal malformations? o0060
A. Tacrolimus o0065
B. Azathioprine o0070
C. Corticosteroids o0075
D. Mycophenolate o0080

4. All of the following agents have been commonly used in the treatment of acute rejection *except:* o0085
A. Antithymocyte globulin o0090
B. Basiliximab o0095
C. Rituximab o0100
D. Intravenous immunoglobulins o0105

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